Code No: 115EH **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	Draw the Layered structure of Operating system.	[2]
b)	Differentiate between multi programming and multi-processing.	[3]
c)	How are processes managed in LINUX?	[2]
d)	Describe the differences between preemptive scheduler and non-preemptive scheduler.	[3]
e)	Under what circumstances do page faults occurs?	[2]
f)	What is the purpose of paging the page tables?	[3]
g)	Describe about File Allocation Table.	[2]
h)	Mention two important functions of Virtual File System (VFS) Layer.	[3]
i)	What is deadlock? What is starvation? How do they differ from each other?	[2]
j)	Write down the principles of protection.	[3]

PART - B

(50 Marks)

2.a)	Define operating system goals from user's view and system's view.	
b)	Explain different categories of System calls with suitable examples.	[5+5]
	OR	
3.a)	Define Essential properties of Distributed Operating Systems.	
b)	Explain about the structure of operating system.	[5+5]
4.	Explain RR and SJF scheduling algorithm with illustrations.	[10]
	OR	
5.	What is a monitor? Explain how dinning philosopher's problem is solved using monitors with example pseudo code.	[10]
6.a)	What is paging? Explain the hardware support given for paging.	
b)	Differentiate Internal Fragmentation and External Fragmentation.	[5+5]
	OR	

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7. Consider the following page reference strings:
1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6.
How many page faults would occur for the following replacement algorithm, assuming three, four frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each.
a) LRU replacement
b) Optimal replacement. [5+5]

8.a) Explain in detail about different File Allocation methods..
b) Explain in detail about file system structure and implementation. [5+5]

OR

9.a) Explain the various methods for free-space management.
b) Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is: 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the SSTF disk scheduling algorithms? [5+5]

10. Consider a system with three processes and four resources. Resource R1 and R3 with one instance, R2 with two instance, process P1 holding an instance of R2 and waiting for R1, process P2 is holding an instance of R1 and R2 and waiting for R3, process P3 is holding an instance of R3.
a) Draw resource allocation graph to the given system.
b) Is it possible to apply the Resource allocation graph algorithm to avoid deadlock? Explain. [5+5]

OR

11.a) What is meant by authentication? Why simple password protection is the most common authentication scheme in use today?
b) Discuss about revocation of access rights. [5+5]

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